T3 A_-2: full general UV Lagrangian and RGEs

Matchete/RGBeta symbolic output in mathematical notation

A_-2

Additional quartics: 5, additional bilinears: 1, extra Yukawa: y2Conj.
General matching and Wilson coefficients are not computed in this symbolic report.

Notation key

symbol	interaction / definition
g_1	$D_\mu \supset i g_1 Y B_\mu$
g_2	$D_\mu \supset i g_2 T^A W_\mu^A$
g_3	$D_\mu \supset i g_3 t^A G_\mu^A$
Y_u	$-\bar{Q} Y_u \widetilde{H} u + \text{h.c.}$
Y_d	$-\bar{Q} Y_d H d + \text{h.c.}$
Y_e	$-\bar{L} Y_e H e + \text{h.c.}$
y_1	$L F S_1 + \text{h.c.}$
y_2	$L F S_2 + \text{h.c.}$
M_F	$-\bar{F} M_F F$
T	$\text{Tr}(Y_e^\dagger Y_e + 3 Y_u^\dagger Y_u + 3 Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\text{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\text{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2	$m_1^2 S_1^\dagger S_1$
m_2^2	$m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2}\lambda_1 (H^\dagger H)^2$
$\lambda_{S_1}^{(1)}$	$\frac{1}{2}\lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$
$\lambda_{S_2}^{(1)}$	$\frac{1}{2}\lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$
$\lambda_{HS_1}^{(1)}$	$\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$
$\lambda_{HS_2}^{(1)}$	$\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$
$\lambda_{HS_1}^{(A)}$	$\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$
$\lambda_{HS_2}^{(A)}$	$\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$
λ_5	$\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$
$\lambda_{12}^{(1)}$	$\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	$\lambda_{12}^{(\times)} (S_1^\dagger S_2)(S_2^\dagger S_1)$
$\lambda_{S_1}^{(A)}$	$\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1)(S_1^\dagger T^A S_1)$
$\lambda_{S_2}^{(A)}$	$\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2)(S_2^\dagger T^A S_2)$
α	$Y_{S_1} = \alpha/2,\ Y_F = (\alpha+1)/2,\ Y_{S_2} = (\alpha+2)/2$
$C_2(d)$	$(d^2-1)/4$
$T(d)$	$d(d^2-1)/12$
$C_{LLS_1S_2}^{ij;AB}$	$L_i L_j S_1^A S_2^B$
C_{ijab}	$L_i L_j \phi_a \phi_b$
C_5^{ij}	$(L_i^T C L_j) H H$
R_W	$16\pi^2 \beta_{C_5} \supset R_W \lambda_5 C_{LLS_1S_2}$

Complete BSM UV Lagrangian

Matchete BSM Lagrangian, excluding the pure Standard Model contribution. $\mathcal{L}_{\text{BSM}} = \text{Coupling(lambdaT3ExtraM000002I1, \{\}, 0)CG(T3ExtraCGM000002I1, \{Index(d\$1, T3BSMd3), Inde}$

$\text{+Coupling(lambdaT3ExtraQ000004I1, \{\}, 0)Bar(Field(NewScalar2, Scalar, \{Index(d881, T3BSMd3)\}, \{\}))Bar(Field(NewScalar2, Scalar, \{Index(d882, T3BSMd3)\}, \{\}))Bar(Field(NewScalar2, Scalar, \{Index(d883, T3BSMd3)\}, \{\}))Bar(Field(NewScalar2, Scalar, \{Index(d884, T3BSMd3)\}, \{\}))CG(T3ExtraCGQ000004I1, \{Index(d881, T3BSMd3), Index(d882, T3BSMd3), Index(d883, T3BSMd3), Index(d884, T3BSMd3)\})}$

$$+\text{Coupling}(\text{lambdaT3ExtraQ02010111},\{\},0)\text{Bar}(\text{Field}(\text{NewScalar1},\text{Scalar},\{\},\{\}))\text{Bar}(\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3})\},\{\}))\text{Bar}(\text{Field}(H,\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund}))\},\{\}))\text{Bar}(\text{Field}(H,\text{Scalar},\{\text{Index}(\text{d\$\$2},\text{SU2L}(\text{fund}))\},\{\}))\text{CG}(\text{T3ExtraCGQ02010111},\{\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund}))\},\text{Index}(\text{d\$\$2},\text{SU2L}(\text{fund})),\text{Index}(\text{d\$\$1},\text{T3BSMd3}))$$

$$+\text{Coupling}(\text{lambdaT3ExtraQ11002I1},\{\},0)\text{Bar}(\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3})\},\{\}))\text{Bar}(\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$2},\text{T3BSMd3})\},\{\}))\text{Field}(H,\text{Scalar},\{\text{Index}(\text{d\$\$2},\text{SU2L}(\text{fund}))\},\{\})\text{Bar}(\text{Field}(H,\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund}))\},\{\}))\text{CG}(\text{T3ExtraCGQ11002I1},\{\text{Bar}(\text{Index}(\text{d\$\$2},\text{SU2L}(\text{fund}))\},\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund})),\text{Index}(\text{d\$\$1},\text{T3BSMd3}),\text{Index}(\text{d\$\$2},\text{T3BSMd3}))$$

$$+\text{Coupling}(\text{lambdaT3ExtraQ001102I1},\{\},0)\text{Field}(\text{NewScalar1},\text{Scalar},\{\},\{\})\text{Bar}(\text{Field}(\text{NewScalar1},\text{Scalar},\{\},\{\}))\text{CG}(\text{T3ExtraCGQ001102I1},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3}),\text{Index}(\text{d\$\$2},\text{T3BSMd3})\})\text{Bar}(\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3})\},\{\}))\text{Bar}(\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$2},\text{T3BSMd3})\},\{\}))$$

$$+\text{Coupling}(\text{MS1},\{\},0)^2\text{Field}(\text{NewScalar1},\text{Scalar},\{\},\{\})(-\text{Bar}(\text{Field}(\text{NewScalar1},\text{Scalar},\{\},\{\})))$$

$$+\text{Coupling}(\text{lambdaH1Inv1},\{\},0)\text{Field}(\text{NewScalar1},\text{Scalar},\{\},\{\})\text{Bar}(\text{Field}(\text{NewScalar1},\text{Scalar},\{\},\{\}))\text{Field}(H,\text{Scalar},\{\text{Index}(\text{d\$\$2},\text{SU2L}(\text{fund}))\},\{\})\text{CG}(\text{T3HiggsPortal1CGInv1},\{\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund})),\text{Bar}(\text{Index}(\text{d\$\$2},\text{SU2L}(\text{fund})))\})\text{Bar}(\text{Field}(H,\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund}))\},\{\}))$$

$$+\text{Coupling}(\text{lambdaT3},\{\},0)\text{Field}(\text{NewScalar1},\text{Scalar},\{\},\{\})\text{Bar}(\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3})\},\{\}))\text{Field}(H,\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund}))\},\{\})\text{Field}(H,\text{Scalar},\{\text{Index}(\text{d\$\$2},\text{SU2L}(\text{fund}))\},\{\})\text{CG}(\text{T3MixCG},\{\text{Bar}(\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund}))),\text{Bar}(\text{Index}(\text{d\$\$2},\text{SU2L}(\text{fund})))\},\text{Index}(\text{d\$\$1},\text{T3BSMd3}))$$

$$+\frac{1}{2}\text{Coupling}(\text{lambdaS1Inv1},\{\},0)\text{Field}(\text{NewScalar1},\text{Scalar},\{\},\{\})^2\text{Bar}(\text{Field}(\text{NewScalar1},\text{Scalar},\{\},\{\}))^2$$

$$+\text{Field}(\text{NewScalar1},\text{Scalar},\{\},\{\text{Index}(\text{d\$\$1},\text{Lorentz})\})\text{Bar}(\text{Field}(\text{NewScalar1},\text{Scalar},\{\},\{\text{Index}(\text{d\$\$1},\text{Lorentz})\}))$$

$$+\text{Bar}(\text{Coupling}(\text{lambdaT3},\{\},0)\text{Bar}(\text{Field}(\text{NewScalar1},\text{Scalar},\{\},\{\}))\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3})\},\{\})\text{Bar}(\text{Field}(H,\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund}))\},\{\}))\text{Bar}(\text{Field}(H,\text{Scalar},\{\text{Index}(\text{d\$\$2},\text{SU2L}(\text{fund}))\},\{\}))\text{CG}(\text{Bar}(\text{T3MixCG}),\{\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund})),\text{Index}(\text{d\$\$2},\text{SU2L}(\text{fund})),\text{Index}(\text{d\$\$1},\text{T3BSMd3})\})$$

$$+\text{Coupling}(\text{MS2},\{\},0)^2\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3})\},\{\})(-\text{Bar}(\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3})\},\{\})))$$

$$+\text{Bar}(\text{Coupling}(\text{lambdaT3ExtraQ02010111},\{\},0)\text{Field}(\text{NewScalar1},\text{Scalar},\{\},\{\})\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3})\},\{\})\text{Field}(H,\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund}))\},\{\})\text{Field}(H,\text{Scalar},\{\text{Index}(\text{d\$\$2},\text{SU2L}(\text{fund}))\},\{\})\text{CG}(\text{Bar}(\text{T3ExtraCGQ02010111}),\{\text{Bar}(\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund}))),\text{Bar}(\text{Index}(\text{d\$\$2},\text{SU2L}(\text{fund}))),\text{Index}(\text{d\$\$1},\text{T3BSMd3})\})$$

$$+\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3})\},\{\text{Index}(\text{d\$\$1},\text{Lorentz})\})\text{Bar}(\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3})\},\{\text{Index}(\text{d\$\$1},\text{Lorentz})\}))$$

$$+\frac{1}{2}\text{Coupling}(\text{lambdaH2Inv1},\{\},0)\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$2},\text{T3BSMd3})\},\{\})\text{Bar}(\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3})\},\{\}))\text{Field}(H,\text{Scalar},\{\text{Index}(\text{d\$\$2},\text{SU2L}(\text{fund}))\},\{\})\text{Bar}(\text{Field}(H,\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund}))\},\{\}))\text{CG}(\text{T3HiggsPortal2CGInv1},\{\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund})),\text{Bar}(\text{Index}(\text{d\$\$2},\text{SU2L}(\text{fund})))\},\text{Index}(\text{d\$\$1},\text{T3BSMd3}),\text{Index}(\text{d\$\$2},\text{T3BSMd3}))$$

$$+\frac{1}{2}\text{Coupling}(\text{lambdaH2Inv1},\{\},0)\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$2},\text{T3BSMd3})\},\{\})\text{Bar}(\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3})\},\{\}))\text{Field}(H,\text{Scalar},\{\text{Index}(\text{d\$\$2},\text{SU2L}(\text{fund}))\},\{\})\text{Bar}(\text{Field}(H,\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund}))\},\{\}))\text{CG}(\text{T3HiggsPortal2CGInv1},\{\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund})),\text{Bar}(\text{Index}(\text{d\$\$2},\text{SU2L}(\text{fund})))\},\text{Index}(\text{d\$\$2},\text{T3BSMd3}),\text{Index}(\text{d\$\$1},\text{T3BSMd3}))$$

$$+\frac{1}{2}\text{Coupling}(\text{lambdaH2Inv2},\{\},0)\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$2},\text{T3BSMd3})\},\{\})\text{Bar}(\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3})\},\{\}))\text{Field}(H,\text{Scalar},\{\text{Index}(\text{d\$\$2},\text{SU2L}(\text{fund}))\},\{\})\text{Bar}(\text{Field}(H,\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund}))\},\{\}))\text{CG}(\text{T3HiggsPortal2CGInv2},\{\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund})),\text{Bar}(\text{Index}(\text{d\$\$2},\text{SU2L}(\text{fund})))\},\text{Index}(\text{d\$\$1},\text{T3BSMd3}),\text{Index}(\text{d\$\$2},\text{T3BSMd3}))$$

$$+\frac{1}{2}\text{Coupling}(\text{lambdaH2Inv2},\{\},0)\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$2},\text{T3BSMd3})\},\{\})\text{Bar}(\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3})\},\{\}))\text{Field}(H,\text{Scalar},\{\text{Index}(\text{d\$\$2},\text{SU2L}(\text{fund}))\},\{\})\text{Bar}(\text{Field}(H,\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund}))\},\{\}))\text{CG}(\text{T3HiggsPortal2CGInv2},\{\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund})),\text{Bar}(\text{Index}(\text{d\$\$2},\text{SU2L}(\text{fund})))\},\text{Index}(\text{d\$\$2},\text{T3BSMd3}),\text{Index}(\text{d\$\$1},\text{T3BSMd3}))$$

$$+\text{Coupling}(\text{lambda12Inv1},\{\},0)\text{Field}(\text{NewScalar1},\text{Scalar},\{\},\{\})\text{Bar}(\text{Field}(\text{NewScalar1},\text{Scalar},\{\},\{\}))\text{CG}(\text{T3CrossScalarCGInv1},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3}),\text{Index}(\text{d\$\$2},\text{T3BSMd3})\})\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$2},\text{T3BSMd3})\},\{\})\text{Bar}(\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3})\},\{\}))$$

$$+\text{Bar}(\text{Coupling}(\text{lambdaT3ExtraM000002I1},\{\},0)\text{CG}(\text{T3ExtraCGM000002I1},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3}),\text{Index}(\text{d\$\$2},\text{T3BSMd3})\})\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3})\},\{\})\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$2},\text{T3BSMd3})\},\{\})$$

$$+\text{Bar}(\text{Coupling}(\text{lambdaT3ExtraQ11002I1},\{\},0)\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3})\},\{\})\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$2},\text{T3BSMd3})\},\{\})\text{Field}(H,\text{Scalar},\{\text{Index}(\text{d\$\$2},\text{SU2L}(\text{fund}))\},\{\})\text{Bar}(\text{Field}(H,\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund}))\},\{\}))\text{CG}(\text{T3ExtraCGQ11002I1},\{\text{Bar}(\text{Index}(\text{d\$\$2},\text{SU2L}(\text{fund}))),\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund})),\text{Index}(\text{d\$\$1},\text{T3BSMd3}),\text{Index}(\text{d\$\$2},\text{T3BSMd3})\})$$

$$+\text{Bar}(\text{Coupling}(\text{lambdaT3ExtraQ001102I1},\{\},0)\text{Field}(\text{NewScalar1},\text{Scalar},\{\},\{\})\text{Bar}(\text{Field}(\text{NewScalar1},\text{Scalar},\{\},\{\}))\text{CG}(\text{T3ExtraCGQ001102I1},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3}),\text{Index}(\text{d\$\$2},\text{T3BSMd3})\})\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3})\},\{\})\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$2},\text{T3BSMd3})\},\{\})$$

$$+\text{Coupling}(\text{lambdaT3ExtraQ00001I3I1},\{\},0)\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$4},\text{T3BSMd3})\},\{\})\text{Bar}(\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3})\},\{\}))\text{Bar}(\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$2},\text{T3BSMd3})\},\{\}))\text{Bar}(\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$3},\text{T3BSMd3})\},\{\}))\text{CG}(\text{T3ExtraCGQ00001I3I1},\{\text{Index}(\text{d\$\$4},\text{T3BSMd3}),\text{Index}(\text{d\$\$1},\text{T3BSMd3}),\text{Index}(\text{d\$\$2},\text{T3BSMd3}),\text{Index}(\text{d\$\$3},\text{T3BSMd3})\})$$

$$+\frac{1}{2}\text{Coupling}(\text{lambdaS2Inv1},\{\},0)\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$3},\text{T3BSMd3})\},\{\})\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$4},\text{T3BSMd3})\},\{\})\text{Bar}(\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3})\},\{\}))\text{Bar}(\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$2},\text{T3BSMd3})\},\{\}))\text{CG}(\text{T3Scalar2SdfCGInv1},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3}),\text{Index}(\text{d\$\$3},\text{T3BSMd3}),\text{Index}(\text{d\$\$2},\text{T3BSMd3}),\text{Index}(\text{d\$\$4},\text{T3BSMd3})\})$$

$$+\frac{1}{2}\text{Coupling}(\text{lambdaS2Inv2},\{\},0)\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$3},\text{T3BSMd3})\},\{\})\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$4},\text{T3BSMd3})\},\{\})\text{Bar}(\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3})\},\{\}))\text{Bar}(\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$2},\text{T3BSMd3})\},\{\}))\text{CG}(\text{T3Scalar2SdfCGInv2},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3}),\text{Index}(\text{d\$\$3},\text{T3BSMd3}),\text{Index}(\text{d\$\$2},\text{T3BSMd3}),\text{Index}(\text{d\$\$4},\text{T3BSMd3})\})$$

$$+\text{Bar}(\text{Coupling}(\text{lambdaT3ExtraQ00001I3I1},\{\},0)\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$2},\text{T3BSMd3})\},\{\})\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$3},\text{T3BSMd3})\},\{\})\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$4},\text{T3BSMd3})\},\{\})\text{Bar}(\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3})\},\{\}))\text{CG}(\text{T3ExtraCGQ00001I3I1},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3}),\text{Index}(\text{d\$\$2},\text{T3BSMd3}),\text{Index}(\text{d\$\$3},\text{T3BSMd3}),\text{Index}(\text{d\$\$4},\text{T3BSMd3})\})$$

$$+\text{Bar}(\text{Coupling}(\text{lambdaT3ExtraQ00000I4I},\{\},0)\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3})\},\{\})\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$2},\text{T3BSMd3})\},\{\})\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$3},\text{T3BSMd3})\},\{\})\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$4},\text{T3BSMd3})\},\{\})\text{CG}(\text{T3ExtraCGQ00000I4I},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3}),\text{Index}(\text{d\$\$2},\text{T3BSMd3}),\text{Index}(\text{d\$\$3},\text{T3BSMd3}),\text{Index}(\text{d\$\$4},\text{T3BSMd3})\})$$

$$-\text{Coupling}(\text{MF},\{\},0)\text{NCM}(\text{Bar}(\text{Field}(\text{NewFermion},\text{Fermion},\{\text{Index}(\text{d\$\$1},\text{T3BSMd2}),\text{Index}(\text{d\$\$1},\text{NFlavor})\},\{\})),\text{Field}(\text{NewFermion},\text{Fermion},\{\text{Index}(\text{d\$\$1},\text{T3BSMd2}),\text{Index}(\text{d\$\$1},\text{NFlavor})\},\{\}))$$

$$+\text{Coupling}(\text{y2},\{\text{Index}(\text{d\$\$1},\text{Flavor}),\text{Index}(\text{d\$\$1},\text{NFlavor})\},0)\text{Bar}(\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3})\},\{\}))\text{CG}(\text{T3Y2CG},\{\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund})),\text{Bar}(\text{Index}(\text{d\$\$1},\text{T3BSMd2})),\text{Index}(\text{d\$\$1},\text{T3BSMd3})\})\text{NCM}(\text{Bar}(\text{Field}(I,\text{Fermion},\{\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund})),\text{Index}(\text{d\$\$1},\text{Flavor})\},\{\})),\text{DiracProduct}(\text{Proj}(1)),\text{Field}(\text{NewFermion},\text{Fermion},\{\text{Index}(\text{d\$\$1},\text{T3BSMd2}),\text{Index}(\text{d\$\$1},\text{NFlavor})\},\{\}))$$

$$+\text{Coupling}(\text{y2Conj},\{\text{Index}(\text{d\$\$1},\text{Flavor}),\text{Index}(\text{d\$\$1},\text{NFlavor})\},0)\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3})\},\{\})\text{CG}(\text{T3Y2ConjCG},\{\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund})),\text{Bar}(\text{Index}(\text{d\$\$1},\text{T3BSMd2})),\text{Index}(\text{d\$\$1},\text{T3BSMd3})\})\text{NCM}(\text{Bar}(\text{Field}(I,\text{Fermion},\{\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund})),\text{Index}(\text{d\$\$1},\text{Flavor})\},\{\})),\text{DiracProduct}(\text{Proj}(1)),\text{Field}(\text{NewFermion},\text{Fermion},\{\text{Index}(\text{d\$\$1},\text{T3BSMd2}),\text{Index}(\text{d\$\$1},\text{NFlavor})\},\{\}))$$

$$+\text{Field}(\text{NewScalar1},\text{Scalar},\{\},\{\})\text{Coupling}(\text{y1},\{\text{Index}(\text{d\$\$1},\text{Flavor}),\text{Index}(\text{d\$\$1},\text{NFlavor})\},0)\text{CG}(\text{T3Y1CG},\{\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund})),\text{Index}(\text{d\$\$1},\text{T3BSMd2})\})\text{NCM}(\text{Bar}(\text{Field}(I,\text{Fermion},\{\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund})),\text{Index}(\text{d\$\$1},\text{Flavor})\},\{\})),\text{DiracProduct}(\text{GammaCC},\text{Proj}(1)),\text{Transp}(\text{Bar}(\text{Field}(\text{NewFermion},\text{Fermion},\{\text{Index}(\text{d\$\$1},\text{T3BSMd2}),\text{Index}(\text{d\$\$1},\text{NFlavor})\},\{\})))$$

$$+i\text{NCM}(\text{Bar}(\text{Field}(\text{NewFermion},\text{Fermion},\{\text{Index}(\text{d\$\$1},\text{T3BSMd2}),\text{Index}(\text{d\$\$1},\text{NFlavor})\},\{\})),\text{DiracProduct}(\text{GammaM}(\text{Index}(\text{d\$\$1},\text{Lorentz}))),\text{Field}(\text{NewFermion},\text{Fermion},\{\text{Index}(\text{d\$\$1},\text{T3BSMd2}),\text{Index}(\text{d\$\$1},\text{NFlavor})\},\{\text{Index}(\text{d\$\$1},\text{Lorentz})\}))$$

$$+\text{Bar}(\text{Coupling}(\text{y2Conj},\{\text{Index}(\text{d\$\$1},\text{Flavor}),\text{Index}(\text{d\$\$1},\text{NFlavor})\},0)\text{Bar}(\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3})\},\{\}))\text{CG}(\text{Bar}(\text{T3Y2ConjCG}),\{\text{Bar}(\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund}))),\text{Index}(\text{d\$\$1},\text{T3BSMd2}),\text{Index}(\text{d\$\$1},\text{T3BSMd3})\})\text{NCM}(\text{Bar}(\text{Field}(\text{NewFermion},\text{Fermion},\{\text{Index}(\text{d\$\$1},\text{T3BSMd2}),\text{Index}(\text{d\$\$1},\text{NFlavor})\},\{\})),\text{DiracProduct}(\text{Proj}(-1)),\text{Field}(I,\text{Fermion},\{\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund})),\text{Index}(\text{d\$\$1},\text{Flavor})\},\{\}))$$

$$+\text{Field}(\text{NewScalar2},\text{Scalar},\{\text{Index}(\text{d\$\$1},\text{T3BSMd3})\},\{\})\text{Bar}(\text{Coupling}(\text{y2},\{\text{Index}(\text{d\$\$1},\text{Flavor}),\text{Index}(\text{d\$\$1},\text{NFlavor})\},0)\text{CG}(\text{Bar}(\text{T3Y2CG}),\{\text{Bar}(\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund}))),\text{Index}(\text{d\$\$1},\text{T3BSMd2}),\text{Index}(\text{d\$\$1},\text{T3BSMd3})\})\text{NCM}(\text{Bar}(\text{Field}(\text{NewFermion},\text{Fermion},\{\text{Index}(\text{d\$\$1},\text{T3BSMd2}),\text{Index}(\text{d\$\$1},\text{NFlavor})\},\{\})),\text{DiracProduct}(\text{Proj}(-1)),\text{Field}(I,\text{Fermion},\{\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund})),\text{Index}(\text{d\$\$1},\text{Flavor})\},\{\}))$$

$$+\text{Bar}(\text{Field}(\text{NewScalar1},\text{Scalar},\{\},\{\}))\text{Bar}(\text{Coupling}(\text{y1},\{\text{Index}(\text{d\$\$1},\text{Flavor}),\text{Index}(\text{d\$\$1},\text{NFlavor})\},0)\text{CG}(\text{Bar}(\text{T3Y1CG}),\{\text{Bar}(\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund}))),\text{Bar}(\text{Index}(\text{d\$\$1},\text{T3BSMd2}))\})\text{NCM}(\text{Transp}(\text{Field}(I,\text{Fermion},\{\text{Index}(\text{d\$\$1},\text{SU2L}(\text{fund})),\text{Index}(\text{d\$\$1},\text{Flavor})\},\{\})),\text{DiracProduct}(\text{GammaCC},\text{Proj}(-1)),\text{Field}(\text{NewFermion},\text{Fermion},\{\text{Index}(\text{d\$\$1},\text{T3BSMd2}),\text{Index}(\text{d\$\$1},\text{NFlavor})\},\{\}))$$

Interaction inventory

- `Yukawa1_GenericCG`
- `Yukawa2_GenericCG`
- `Yukawa2ConjugateScalar_GenericCG`
- `Scalar1SelfInv1`
- `Scalar2SelfInv1`
- `Scalar2SelfInv2`
- `HiggsPortal1Inv1`
- `HiggsPortal2Inv1`
- `HiggsPortal2Inv2`
- `ScalarCrossPortalInv1`

- T3ScalarMix_GenericCG
- T3ExtraQ_0_0_0_0_0_4_1
- T3ExtraQ_0_0_0_0_1_3_1
- T3ExtraQ_0_0_1_1_0_2_1
- T3ExtraQ_0_2_0_1_0_1_1
- T3ExtraQ_1_1_0_0_0_2_1
- T3ExtraM_0_0_0_0_0_2_1

One-loop renormalisable beta functions

$16\pi^2\, dX/d\ln\mu = \beta_X$

g_1

$$16\pi^2 \frac{dg_1}{d\ln\mu} = \frac{55{g_1}^3}{6}$$

g_2

$$16\pi^2 \frac{dg_2}{d\ln\mu} = -\frac{{g_2}^3}{2}$$

g_3

$$16\pi^2 \frac{dg_3}{d\ln\mu} = -7{g_3}^3$$

Y_u

$$\begin{aligned}
16\pi^2 \frac{dY_u}{d\ln\mu} = {} & 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_u)_{ij} \\
& + \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_u)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_u)_{ij} \\
& - \frac{3}{2}(\Big(Y_d\ (Y_d)^\dagger\ Y_u\Big)_{ij} - \Big(Y_u\ (Y_u)^\dagger\ Y_u\Big)_{ij}) \\
& - \frac{9}{4}{g_2}^2\,(Y_u)_{ij} \\
& - 8{g_3}^2\,(Y_u)_{ij} \\
& - \frac{17}{12}{g_1}^2\,(Y_u)_{ij}
\end{aligned}$$

Y_d

$$\begin{aligned}
16\pi^2 \frac{dY_d}{d\ln\mu} = {} & \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij} \\
& + \frac{3}{2}(\Big(Y_d\ (Y_d)^\dagger\ Y_d\Big)_{ij} - \Big(Y_u\ (Y_u)^\dagger\ Y_d\Big)_{ij}) \\
& - \frac{9}{4}{g_2}^2\,(Y_d)_{ij} \\
& - 8{g_3}^2\,(Y_d)_{ij} \\
& - \frac{5}{12}{g_1}^2\,(Y_d)_{ij}
\end{aligned}$$

Y_e

$$\begin{aligned}16\pi^2 \frac{dY_e}{d\ln\mu} &= 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_e)_{ij} \\ &+ 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_e)_{ij} \\ &+ \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_e)_{ij} \\ &+ \frac{1}{4}(2\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij} + 3(\left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij} + \left(\mathrm{y2Conj}\ (\mathrm{y2Conj})^\dagger\ Y_e\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})) \\ &- \frac{9}{4}{g_2}^2\,(Y_e)_{ij} \\ &- \frac{15}{4}{g_1}^2\,(Y_e)_{ij}\end{aligned}$$

y_1

$$\begin{aligned}16\pi^2 \frac{dy_1}{d\ln\mu} &= 2T_\nu^{(1)}\,(y_1)_{ij} \\ &+ \frac{1}{4}(3\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij} + 3\left(\mathrm{y2Conj}\ (\mathrm{y2Conj})^\dagger\ y_1\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij} + 4\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij}) \\ &- \frac{9}{2}{g_2}^2\,(y_1)_{ij} \\ &- \frac{3}{2}{g_1}^2\,(y_1)_{ij}\end{aligned}$$

y_2

$$\begin{aligned}16\pi^2 \frac{dy_2}{d\ln\mu} &= \mathrm{Tr}\Big(\mathrm{y2Conj}\ (\mathrm{y2Conj})^\dagger\Big)\,(y_2)_{ij} \\ &+ 2\,\mathrm{Tr}\Big(y_2\ (\mathrm{y2Conj})^\dagger\Big)\,(\mathrm{y2Conj})_{ij} \\ &+ T_\nu^{(2)}\,(y_2)_{ij} \\ &+ \frac{1}{12}(6\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij} + 29\left(y_2\ (\mathrm{y2Conj})^\dagger\ \mathrm{y2Conj}\right)_{ij} + 29\left(\mathrm{y2Conj}\ (\mathrm{y2Conj})^\dagger\ y_2\right)_{ij} + 6\left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij} + 18\Big(\\ &- \frac{9}{2}{g_2}^2\,(y_2)_{ij} \\ &- \frac{3}{2}{g_1}^2\,(y_2)_{ij}\end{aligned}$$

M_F

$$\begin{aligned}16\pi^2 \frac{dM_F}{d\ln\mu} &= \frac{1}{4}(2\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij} + 3\left(M_F\ (y_2)^\dagger\ y_2\right)_{ij} + 3\left(M_F\ (\mathrm{y2Conj})^\dagger\ \mathrm{y2Conj}\right)_{ij}) \\ &- \frac{9}{2}{g_2}^2\,(M_F)_{ij} \\ &- \frac{3}{2}{g_1}^2\,(M_F)_{ij}\end{aligned}$$

m_1^2

$$16\pi^2 \frac{dm_1^2}{d\ln\mu} = m_1^2(4T_\nu^{(1)} + 4\lambda_{S_1}^{(1)})$$

$$
\begin{aligned}
&- 8\,\mathrm{Tr}\Big(M_F\ (M_F)^\dagger\ (y_1)^T\ (y_1)^*\Big)\\
&+ 12\mathrm{muT3GeneralS2Sq}\,(\mathrm{lambdaT3GeneralQ001102I1})^*\\
&+ 12\mathrm{lambdaT3GeneralQ001102I1}\,(\mathrm{muT3GeneralS2Sq})^*\\
&- 6{g_1}^2 m_1^2\\
&+ 6\lambda_{12}^{(1)} m_2^2
\end{aligned}
$$

m_2^2

$$
\begin{aligned}
16\pi^2\,\frac{dm_2^2}{d\ln\mu} = {}& m_2^2(2T_\nu^{(2)} + 2\,\mathrm{Tr}\Big(\mathrm{y2Conj}\ (\mathrm{y2Conj})^\dagger\Big) + 8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})\\
&+ \mathrm{muT3GeneralS2Sq}(4\,\mathrm{Tr}\Big(\mathrm{y2Conj}\ (y_2)^\dagger\Big) + 20\,(\mathrm{lambdaT3GeneralQ000013I1})^*)\\
&- 4\,\mathrm{Tr}\Big(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\Big)\\
&- 4\,\mathrm{Tr}\Big(M_F\ (\mathrm{y2Conj})^\dagger\ \mathrm{y2Conj}\ (M_F)^\dagger\Big)\\
&+ 4\,(\mathrm{muT3GeneralS2Sq})^*\,\mathrm{Tr}\Big(y_2\ (\mathrm{y2Conj})^\dagger\Big)\\
&+ 20\mathrm{lambdaT3GeneralQ000013I1}\,(\mathrm{muT3GeneralS2Sq})^*\\
&- 12{g_2}^2 m_2^2\\
&+ 2\lambda_{12}^{(1)} m_1^2
\end{aligned}
$$

λ_1

$$
\begin{aligned}
16\pi^2\,\frac{d\lambda_1}{d\ln\mu} = {}& \lambda_1(12\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 4\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 12\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big))\\
&- 12\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\Big)\\
&- 4\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\Big)\\
&- 12\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\Big)\\
&+ 4\lambda_5\,(\lambda_5)^*\\
&+ 4\mathrm{lambdaT3GeneralQ020101I1}\,(\mathrm{lambdaT3GeneralQ020101I1})^*\\
&+ 24\mathrm{lambdaT3GeneralQ110002I1}\,(\mathrm{lambdaT3GeneralQ110002I1})^*\\
&+ \frac{9{g_2}^4}{4}\\
&+ {g_1}^2\left(\frac{3{g_2}^2}{2} - 3\lambda_1\right)\\
&- 9{g_2}^2\lambda_1\\
&+ \frac{3{g_1}^4}{4}\\
&+ 12{\lambda_1}^2\\
&+ 2{\lambda_{HS_1}^{(1)}}^2\\
&+ 6{\lambda_{HS_2}^{(1)}}^2\\
&+ 3{\lambda_{HS_2}^{(A)}}^2
\end{aligned}
$$

$\lambda_{S_1}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{S_1}^{(1)}}{d\ln\mu} &= 8\lambda_{S_1}^{(1)} T_\nu^{(1)} \\
&\quad - 8\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\Big) \\
&\quad + 24\mathrm{lambdaT3GeneralQ001102I1}\ (\mathrm{lambdaT3GeneralQ001102I1})^* \\
&\quad + 12{g_1}^4 \\
&\quad - 12{g_1}^2\lambda_{S_1}^{(1)} \\
&\quad + 6{\lambda_{12}^{(1)}}^2 \\
&\quad + 4{\lambda_{HS_1}^{(1)}}^2 \\
&\quad + 10{\lambda_{S_1}^{(1)}}^2
\end{aligned}
$$

$\lambda_{S_2}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{S_2}^{(1)}}{d\ln\mu} &= \lambda_{S_2}^{(1)}(4T_\nu^{(2)} + 4\,\mathrm{Tr}\Big(\mathrm{y2Conj}\ (\mathrm{y2Conj})^\dagger\Big) + 8\lambda_{S_2}^{(A)}) \\
&\quad + \frac{32}{3}\ (\mathrm{lambdaT3GeneralQ000013I1})^*\,\mathrm{Tr}\Big(y_2\ (\mathrm{y2Conj})^\dagger\Big) \\
&\quad + \mathrm{lambdaT3GeneralQ000013I1}\left(\frac{32}{3}\,\mathrm{Tr}\Big(\mathrm{y2Conj}\ (y_2)^\dagger\Big) + \frac{352\,(\mathrm{lambdaT3GeneralQ000013I1})^*}{3}\right) \\
&\quad - \frac{248}{45}\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ \mathrm{y2Conj}\ (\mathrm{y2Conj})^\dagger\Big) \\
&\quad - \frac{248}{45}\,\mathrm{Tr}\Big(y_2\ (\mathrm{y2Conj})^\dagger\ \mathrm{y2Conj}\ (y_2)^\dagger\Big) \\
&\quad - \frac{8}{3}\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big) \\
&\quad - \frac{8}{3}\,\mathrm{Tr}\Big(\mathrm{y2Conj}\ (\mathrm{y2Conj})^\dagger\ \mathrm{y2Conj}\ (\mathrm{y2Conj})^\dagger\Big) \\
&\quad + \frac{832}{3}\mathrm{lambdaT3GeneralQ000004I1}\ (\mathrm{lambdaT3GeneralQ000004I1})^* \\
&\quad + \frac{8}{3}\mathrm{lambdaT3GeneralQ001102I1}\ (\mathrm{lambdaT3GeneralQ001102I1})^* \\
&\quad + \frac{16}{3}\mathrm{lambdaT3GeneralQ110002I1}\ (\mathrm{lambdaT3GeneralQ110002I1})^* \\
&\quad + 22{g_2}^4 \\
&\quad - 24{g_2}^2\lambda_{S_2}^{(1)} \\
&\quad + 2{\lambda_{12}^{(1)}}^2 \\
&\quad + 4{\lambda_{HS_2}^{(1)}}^2 \\
&\quad + 14{\lambda_{S_2}^{(1)}}^2 \\
&\quad + 16{\lambda_{S_2}^{(A)}}^2
\end{aligned}
$$

$\lambda_{HS_1}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_1}^{(1)}}{d\ln\mu} &= \lambda_{HS_1}^{(1)}(4T_\nu^{(1)} + 6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)) \\
&\quad - 4\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\Big)
\end{aligned}
$$

$$
\begin{aligned}
&+ 12\lambda_5 \left(\lambda_5\right)^* \\
&+ 12\text{lambdaT3GeneralQ110002I1} \left(\text{lambdaT3GeneralQ001102I1}\right)^* \\
&+ 12\text{lambdaT3GeneralQ001102I1} \left(\text{lambdaT3GeneralQ110002I1}\right)^* \\
&+ 12\text{lambdaT3GeneralQ020101I1} \left(\text{lambdaT3GeneralQ020101I1}\right)^* \\
&- \frac{9{g_2}^2\lambda_{HS_1}^{(1)}}{2} \\
&+ 3{g_1}^4 \\
&- \frac{15{g_1}^2\lambda_{HS_1}^{(1)}}{2} \\
&+ 6\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)} \\
&+ 6\lambda_1\lambda_{HS_1}^{(1)} \\
&+ 4{\lambda_{HS_1}^{(1)}}^2 \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}
\end{aligned}
$$

$\lambda_{HS_2}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_2}^{(1)}}{d\ln\mu} = {}& \lambda_{HS_2}^{(1)}(2T_\nu^{(2)} + 2\,\mathrm{Tr}\left(\text{y2Conj}\ \left(\text{y2Conj}\right)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_d\ \left(Y_d\right)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ \left(Y_e\right)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ \left(Y_u\right)^\dagger\right) + 4\lambda_{S_2}^{(A)}) \\
&+ \text{lambdaT3GeneralQ110002I1}(4\,\mathrm{Tr}\left(\text{y2Conj}\ \left(y_2\right)^\dagger\right) + 4(5\left(\text{lambdaT3GeneralQ000013I1}\right)^* + 4\left(\text{lambdaT3Gene}\right. \\
&+ 4\left(\text{lambdaT3GeneralQ110002I1}\right)^*\mathrm{Tr}\left(y_2\ \left(\text{y2Conj}\right)^\dagger\right) \\
&- 2\,\mathrm{Tr}\left(y_2\ \left(y_2\right)^\dagger\ Y_e\ \left(Y_e\right)^\dagger\right) \\
&- 2\,\mathrm{Tr}\left(\text{y2Conj}\ \left(\text{y2Conj}\right)^\dagger\ Y_e\ \left(Y_e\right)^\dagger\right) \\
&+ 4\lambda_5 \left(\lambda_5\right)^* \\
&+ 20\text{lambdaT3GeneralQ000013I1} \left(\text{lambdaT3GeneralQ110002I1}\right)^* \\
&+ 4\text{lambdaT3GeneralQ020101I1} \left(\text{lambdaT3GeneralQ020101I1}\right)^* \\
&+ 6{g_2}^4 \\
&- \frac{33{g_2}^2\lambda_{HS_2}^{(1)}}{2} \\
&- \frac{3{g_1}^2\lambda_{HS_2}^{(1)}}{2} \\
&+ 2\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)} \\
&+ 6\lambda_1\lambda_{HS_2}^{(1)} \\
&+ 4{\lambda_{HS_2}^{(1)}}^2 \\
&+ 8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)} \\
&+ 6{\lambda_{HS_2}^{(A)}}^2
\end{aligned}
$$

$\lambda_{12}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{12}^{(1)}}{d\ln\mu} = {}& \lambda_{12}^{(1)}(4T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}\left(\text{y2Conj}\ \left(\text{y2Conj}\right)^\dagger\right) + 4\lambda_{S_2}^{(A)}) \\
&+ \text{lambdaT3GeneralQ001102I1}(4\,\mathrm{Tr}\left(\text{y2Conj}\ \left(y_2\right)^\dagger\right) + 4(5\left(\text{lambdaT3GeneralQ000013I1}\right)^* + 4\left(\text{lambdaT3Gener}\right. \\
&+ 4\left(\text{lambdaT3GeneralQ001102I1}\right)^*\mathrm{Tr}\left(y_2\ \left(\text{y2Conj}\right)^\dagger\right)
\end{aligned}
$$

$$
\begin{aligned}
&- 4\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)\\
&- 4\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ \mathrm{y2Conj}\ (\mathrm{y2Conj})^\dagger\Big)\\
&+ 4\lambda_5\,(\lambda_5)^*\\
&+ 20\mathrm{lambdaT3GeneralQ000013I1}\,(\mathrm{lambdaT3GeneralQ001102I1})^*\\
&+ 4\mathrm{lambdaT3GeneralQ020101I1}\,(\mathrm{lambdaT3GeneralQ020101I1})^*\\
&- 12{g_2}^2\lambda_{12}^{(1)}\\
&- 6{g_1}^2\lambda_{12}^{(1)}\\
&+ 4{\lambda_{12}^{(1)}}^2\\
&+ 4\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}\\
&+ 8\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}\\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}
\end{aligned}
$$

λ_5

$$
\begin{aligned}
16\pi^2\,\frac{d\lambda_5}{d\ln\mu} = {}&\lambda_5(2T_\nu^{(1)} + T_\nu^{(2)} + \mathrm{Tr}\Big(\mathrm{y2Conj}\ (\mathrm{y2Conj})^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) + 4\lambda_{HS_2}^{(A)})\\
&+ 2\,(\mathrm{lambdaT3GeneralQ020101I1})^*\,\mathrm{Tr}\Big(y_2\ (\mathrm{y2Conj})^\dagger\Big)\\
&+ 4\mathrm{lambdaT3GeneralQ001102I1}\,(\mathrm{lambdaT3GeneralQ020101I1})^*\\
&+ 8\mathrm{lambdaT3GeneralQ110002I1}\,(\mathrm{lambdaT3GeneralQ020101I1})^*\\
&- \frac{21{g_2}^2\lambda_5}{2}\\
&- \frac{9{g_1}^2\lambda_5}{2}\\
&+ 2\lambda_{12}^{(1)}\lambda_5\\
&+ 2\lambda_1\lambda_5\\
&+ 4\lambda_{HS_1}^{(1)}\lambda_5\\
&+ 4\lambda_{HS_2}^{(1)}\lambda_5
\end{aligned}
$$

$\lambda_{HS_2}^{(A)}$

$$
\begin{aligned}
16\pi^2\,\frac{d\lambda_{HS_2}^{(A)}}{d\ln\mu} = {}&\lambda_{HS_2}^{(A)}(2T_\nu^{(2)} + 2\,\mathrm{Tr}\Big(\mathrm{y2Conj}\ (\mathrm{y2Conj})^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) + 16\lambda_{S_2}^{(A)})\\
&+ \frac{4}{3}\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\Big)\\
&- \frac{4}{3}\,\mathrm{Tr}\Big(\mathrm{y2Conj}\ (\mathrm{y2Conj})^\dagger\ Y_e\ (Y_e)^\dagger\Big)\\
&+ \frac{8}{3}\lambda_5\,(\lambda_5)^*\\
&- \frac{8}{3}\mathrm{lambdaT3GeneralQ020101I1}\,(\mathrm{lambdaT3GeneralQ020101I1})^*\\
&- \frac{33{g_2}^2\lambda_{HS_2}^{(A)}}{2}\\
&- \frac{3{g_1}^2\lambda_{HS_2}^{(A)}}{2}\\
&+ 2\lambda_1\lambda_{HS_2}^{(A)}\\
&+ 8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}
\end{aligned}
$$

$$+ 2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$$

$\lambda_{S_2}^{(A)}$

$$\begin{aligned}
16\pi^2 \frac{d\lambda_{S_2}^{(A)}}{d\ln\mu} &= \lambda_{S_2}^{(A)}(4T_\nu^{(2)} + 4\,\mathrm{Tr}\Big(\mathrm{y2Conj}\ (\mathrm{y2Conj})^\dagger\Big)) \\
&\quad - \frac{4}{3}\left(\mathrm{lambdaT3GeneralQ000013I1}\right)^* \mathrm{Tr}\Big(y_2\ (\mathrm{y2Conj})^\dagger\Big) \\
&\quad + \mathrm{lambdaT3GeneralQ000013I1}\left(-\frac{4}{3}\,\mathrm{Tr}\Big(\mathrm{y2Conj}\ (y_2)^\dagger\Big) - \frac{44\left(\mathrm{lambdaT3GeneralQ000013I1}\right)^*}{3}\right) \\
&\quad + \frac{76}{45}\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ \mathrm{y2Conj}\ (\mathrm{y2Conj})^\dagger\Big) \\
&\quad + \frac{76}{45}\,\mathrm{Tr}\Big(y_2\ (\mathrm{y2Conj})^\dagger\ \mathrm{y2Conj}\ (y_2)^\dagger\Big) \\
&\quad - \frac{2}{3}\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big) \\
&\quad - \frac{2}{3}\,\mathrm{Tr}\Big(\mathrm{y2Conj}\ (\mathrm{y2Conj})^\dagger\ \mathrm{y2Conj}\ (\mathrm{y2Conj})^\dagger\Big) \\
&\quad - \frac{224}{3}\mathrm{lambdaT3GeneralQ000004I1}\left(\mathrm{lambdaT3GeneralQ000004I1}\right)^* \\
&\quad - \frac{4}{3}\mathrm{lambdaT3GeneralQ001102I1}\left(\mathrm{lambdaT3GeneralQ001102I1}\right)^* \\
&\quad - \frac{8}{3}\mathrm{lambdaT3GeneralQ110002I1}\left(\mathrm{lambdaT3GeneralQ110002I1}\right)^* \\
&\quad - 2{g_2}^4 \\
&\quad - 24{g_2}^2\lambda_{S_2}^{(A)} \\
&\quad + 3{\lambda_{HS_2}^{(A)}}^2 \\
&\quad + 12\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)} \\
&\quad + 20{\lambda_{S_2}^{(A)}}^2
\end{aligned}$$

$y2Conj$

$$\begin{aligned}
16\pi^2 \frac{dy2Conj}{d\ln\mu} &= 2\,\mathrm{Tr}\Big(\mathrm{y2Conj}\ (y_2)^\dagger\Big)\,(y_2)_{ij} \\
&\quad + T_\nu^{(2)}\,(\mathrm{y2Conj})_{ij} \\
&\quad + \mathrm{Tr}\Big(\mathrm{y2Conj}\ (\mathrm{y2Conj})^\dagger\Big)\,(\mathrm{y2Conj})_{ij} \\
&\quad + \frac{1}{12}(6\left(y_1\ (y_1)^\dagger\ \mathrm{y2Conj}\right)_{ij} + 29\left(y_2\ (y_2)^\dagger\ \mathrm{y2Conj}\right)_{ij} + 29\left(\mathrm{y2Conj}\ (y_2)^\dagger\ y_2\right)_{ij} + 6\left(Y_e\ (Y_e)^\dagger\ \mathrm{y2Conj}\right)_{ij} + \\
&\quad - \frac{9}{2}{g_2}^2\,(\mathrm{y2Conj})_{ij} \\
&\quad - \frac{3}{2}{g_1}^2\,(\mathrm{y2Conj})_{ij}
\end{aligned}$$

$lambdaT3GeneralQ000004I1$

$$\begin{aligned}
16\pi^2 \frac{dlambdaT3GeneralQ000004I1}{d\ln\mu} &= \mathrm{lambdaT3GeneralQ000004I1}(4T_\nu^{(2)} + 4\,\mathrm{Tr}\Big(\mathrm{y2Conj}\ (\mathrm{y2Conj})^\dagger\Big)) \\
&\quad + 2\mathrm{lambdaT3GeneralQ000013I1}\,\mathrm{Tr}\Big(y_2\ (\mathrm{y2Conj})^\dagger\Big)
\end{aligned}$$

$$
\begin{aligned}
&- \frac{23}{15}\,\mathrm{Tr}\Big(y_2\ (\mathrm{y2Conj})^\dagger\ y_2\ (\mathrm{y2Conj})^\dagger\Big)\\
&- 24{g_2}^2\mathrm{lambdaT3GeneralQ000004I1}\\
&+ 12\lambda_{S_2}^{(1)}\mathrm{lambdaT3GeneralQ000004I1}\\
&- 8\lambda_{S_2}^{(A)}\mathrm{lambdaT3GeneralQ000004I1}\\
&+ 11\mathrm{lambdaT3GeneralQ000013I1}^2\\
&+ \mathrm{lambdaT3GeneralQ001102I1}^2\\
&+ 2\mathrm{lambdaT3GeneralQ110002I1}^2
\end{aligned}
$$

$lambdaT3GeneralQ000013I1$

$$
\begin{aligned}
16\pi^2\,\frac{dlambdaT3GeneralQ000013I1}{d\ln\mu} &= \lambda_{S_2}^{(1)}(2\,\mathrm{Tr}\Big(y_2\ (\mathrm{y2Conj})^\dagger\Big) + 22\mathrm{lambdaT3GeneralQ000013I1})\\
&+ \mathrm{lambdaT3GeneralQ000004I1}(8\,\mathrm{Tr}\Big(\mathrm{y2Conj}\ (y_2)^\dagger\Big) + 88\,(\mathrm{lambdaT3GeneralQ000013I}\\
&+ \mathrm{lambdaT3GeneralQ000013I1}(4T_\nu^{(2)} + 4\,\mathrm{Tr}\Big(\mathrm{y2Conj}\ (\mathrm{y2Conj})^\dagger\Big))\\
&- \frac{46}{15}\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (\mathrm{y2Conj})^\dagger\Big)\\
&- \frac{46}{15}\,\mathrm{Tr}\Big(y_2\ (\mathrm{y2Conj})^\dagger\ \mathrm{y2Conj}\ (\mathrm{y2Conj})^\dagger\Big)\\
&- 24{g_2}^2\mathrm{lambdaT3GeneralQ000013I1}\\
&+ 2\lambda_{12}^{(1)}\mathrm{lambdaT3GeneralQ001102I1}\\
&+ 4\lambda_{HS_2}^{(1)}\mathrm{lambdaT3GeneralQ110002I1}
\end{aligned}
$$

$lambdaT3GeneralQ001102I1$

$$
\begin{aligned}
16\pi^2\,\frac{dlambdaT3GeneralQ001102I1}{d\ln\mu} &= \lambda_{12}^{(1)}(2\,\mathrm{Tr}\Big(y_2\ (\mathrm{y2Conj})^\dagger\Big) + 10\mathrm{lambdaT3GeneralQ000013I1} + 8\mathrm{lambdaT3GeneralQ001}\\
&+ \mathrm{lambdaT3GeneralQ001102I1}(4T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}\Big(\mathrm{y2Conj}\ (\mathrm{y2Conj})^\dagger\Big))\\
&- 4\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (\mathrm{y2Conj})^\dagger\Big)\\
&+ 40\mathrm{lambdaT3GeneralQ000004I1}\,(\mathrm{lambdaT3GeneralQ001102I1})^*\\
&- 12{g_2}^2\mathrm{lambdaT3GeneralQ001102I1}\\
&- 6{g_1}^2\mathrm{lambdaT3GeneralQ001102I1}\\
&+ 4\lambda_{HS_1}^{(1)}\mathrm{lambdaT3GeneralQ110002I1}\\
&+ 4\lambda_{S_1}^{(1)}\mathrm{lambdaT3GeneralQ001102I1}\\
&+ 2\lambda_{S_2}^{(1)}\mathrm{lambdaT3GeneralQ001102I1}\\
&- 4\lambda_{S_2}^{(A)}\mathrm{lambdaT3GeneralQ001102I1}\\
&+ 4\lambda_5\mathrm{lambdaT3GeneralQ020101I1}
\end{aligned}
$$

$lambdaT3GeneralQ020101I1$

$$
\begin{aligned}
16\pi^2\,\frac{dlambdaT3GeneralQ020101I1}{d\ln\mu} &= 2\,(\lambda_5)^*\,\mathrm{Tr}\Big(y_2\ (\mathrm{y2Conj})^\dagger\Big)\\
&+ \mathrm{lambdaT3GeneralQ020101I1}(2T_\nu^{(1)} + T_\nu^{(2)} + \mathrm{Tr}\Big(\mathrm{y2Conj}\ (\mathrm{y2Conj})^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_d\ (Y_d)\\
&+ 4\mathrm{lambdaT3GeneralQ001102I1}\,(\lambda_5)^*
\end{aligned}
$$

$$
\begin{aligned}
&+ 8\text{lambdaT3GeneralQ110002I1}\,(\lambda_5)^* \\
&- \frac{21{g_2}^2\text{lambdaT3GeneralQ020101I1}}{2} \\
&- \frac{9{g_1}^2\text{lambdaT3GeneralQ020101I1}}{2} \\
&+ 2\lambda_{12}^{(1)}\text{lambdaT3GeneralQ020101I1} \\
&+ 2\lambda_1\text{lambdaT3GeneralQ020101I1} \\
&+ 4\lambda_{HS_1}^{(1)}\text{lambdaT3GeneralQ020101I1} \\
&+ 4\lambda_{HS_2}^{(1)}\text{lambdaT3GeneralQ020101I1} \\
&- 4\lambda_{HS_2}^{(A)}\text{lambdaT3GeneralQ020101I1}
\end{aligned}
$$

$lambdaT3GeneralQ110002I1$

$$
\begin{aligned}
16\pi^2\,\frac{d\,lambdaT3GeneralQ110002I1}{d\ln\mu} &= \lambda_{HS_2}^{(1)}(2\,\mathrm{Tr}\Big(y_2\ (\text{y2Conj})^\dagger\Big) + 10\text{lambdaT3GeneralQ000013I1} + 8\text{lambdaT3GeneralQ11} \\
&+ \text{lambdaT3GeneralQ110002I1}(2T_\nu^{(2)} + 2\,\mathrm{Tr}\Big(\text{y2Conj}\ (\text{y2Conj})^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + \\
&- 2\,\mathrm{Tr}\Big(y_2\ (\text{y2Conj})^\dagger\ Y_e\ (Y_e)^\dagger\Big) \\
&+ 40\text{lambdaT3GeneralQ000004I1}\,(\text{lambdaT3GeneralQ110002I1})^* \\
&- \frac{33{g_2}^2\text{lambdaT3GeneralQ110002I1}}{2} \\
&- \frac{3{g_1}^2\text{lambdaT3GeneralQ110002I1}}{2} \\
&+ 6\lambda_1\text{lambdaT3GeneralQ110002I1} \\
&+ 2\lambda_{HS_1}^{(1)}\text{lambdaT3GeneralQ001102I1} \\
&+ 2\lambda_{S_2}^{(1)}\text{lambdaT3GeneralQ110002I1} \\
&- 4\lambda_{S_2}^{(A)}\text{lambdaT3GeneralQ110002I1} \\
&+ 4\lambda_5\text{lambdaT3GeneralQ020101I1}
\end{aligned}
$$

$muT3GeneralS2Sq$

$$
\begin{aligned}
16\pi^2\,\frac{d\,muT3GeneralS2Sq}{d\ln\mu} &= m_2^2(2\,\mathrm{Tr}\Big(y_2\ (\text{y2Conj})^\dagger\Big) + 10\text{lambdaT3GeneralQ000013I1}) \\
&- 4\,\mathrm{Tr}\Big(M_F\ (\text{y2Conj})^\dagger\ y_2\ (M_F)^\dagger\Big) \\
&+ \text{muT3GeneralS2Sq}(2T_\nu^{(2)} + 2\,\mathrm{Tr}\Big(\text{y2Conj}\ (\text{y2Conj})^\dagger\Big)) \\
&+ 40\text{lambdaT3GeneralQ000004I1}\,(\text{muT3GeneralS2Sq})^* \\
&- 12{g_2}^2\text{muT3GeneralS2Sq} \\
&+ 2\lambda_{S_2}^{(1)}\text{muT3GeneralS2Sq} \\
&- 4\lambda_{S_2}^{(A)}\text{muT3GeneralS2Sq} \\
&+ 2\text{lambdaT3GeneralQ001102I1}m_1^2
\end{aligned}
$$